

- 4 (a) "The magnitude of *gravitational field strength* at a point is proportional to the *gravitational potential gradient* at that point."

Explain the terms in italics.

gravitational field strength: **gravitational force per unit mass**

gravitational potential gradient: **change in gravitational potential energy per unit mass per unit distance, or work done per unit mass against gravitational force per unit distance**

- (b) Fig. 4.1 shows four equipotential lines at 1000-km intervals in a non-uniform gravitational field.

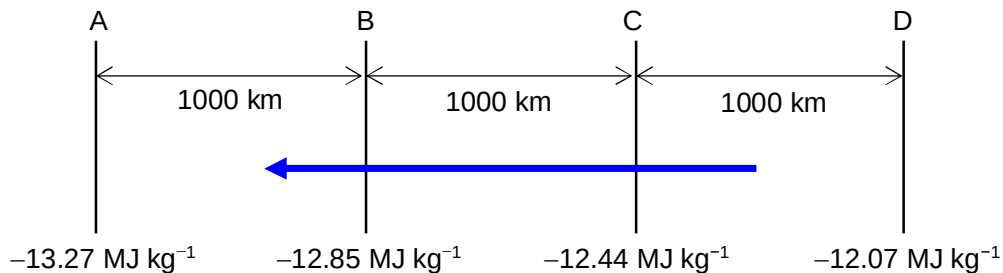


Fig. 4.1

- (i) Draw an arrow on Fig 4.1 to indicate the direction of the gravitational field. [1]
 (ii) Calculate the ratio of

$$\frac{\text{field strength at B}}{\text{field strength at C}}$$

$$\text{Field strength at B} = (13.27 - 12.44) \times 10^6 / 2000 \times 10^3 = 0.415 \text{ N kg}^{-1}$$

$$\text{Field strength at C} = (12.85 - 12.07) \times 10^6 / 2000 \times 10^3 = 0.390 \text{ N kg}^{-1}$$

$$\text{Ratio} = 0.415 / 0.390 = 1.06$$

ratio = [3]

- (c) A theoretical long open tube is placed in the gravitational field as shown in Fig. 4.2. An object is released from rest at B.

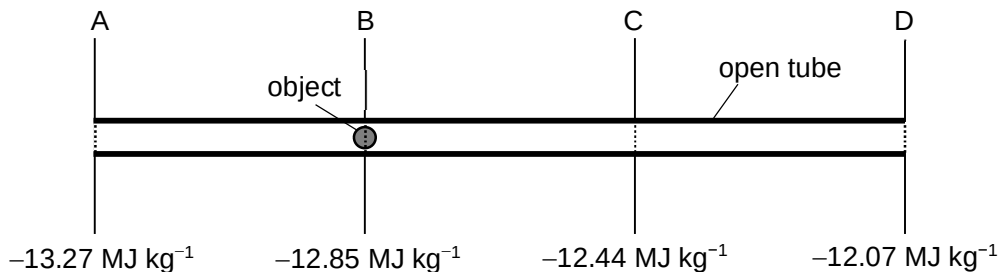


Fig. 4.2

Determine the speed with which the object leaves the tube.

object will accelerate to the left

$$\text{Loss in GPE} = m (13.27 - 12.85) \times 10^6 = 0.42 \times 10^6 \text{ m}$$

$$\text{Gain in KE} = \frac{1}{2} m v^2 - 0 = 0.42 \times 10^6 \text{ m}$$

$$v = 917 \text{ m s}^{-1}$$

speed = m s^{-1} [4]